

RFID-BASED INVENTORY MANAGEMENT SYSTEM FOR A RETAIL STORE

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ABSTRACT

This study implements an RFID-based inventory management system for a small retail store, reducing stockout incidents by 65% and inventory counting time by 80% through automated real-time tracking.

INTRODUCTION

Small retail stores in the Philippines commonly rely on manual inventory count, resulting in stockouts, overstocking, and count errors. RFID technology enables automated real-time item tracking without line-of-sight scanning. This study designs and deploys an RFID inventory system for a grocery store in Imus City with approximately 1,200 SKUs.

METHODOLOGY

RFID tags (MIFARE Classic 1K) were attached to product units. Two RFID readers (RC522 modules) were installed at entry and exit gates, connected to Arduino Mega 2560. A web dashboard built in Flask tracked inventory counts in real time with PostgreSQL storage. Threshold alerts were configured per SKU (reorder point = safety stock + average daily demand $\times$ lead time). The system was deployed and monitored over 90 days.

RESULTS

Stockout incidents decreased from 23 per month to 8 per month (65% reduction). Inventory counting time decreased from 4.5 hours to 53 minutes (80% reduction). Inventory accuracy improved from 91.2% to 98.7%. Reader accuracy was 99.4% at a maximum read distance of 8cm. System uptime was 98.9% over the 90-day evaluation.

DISCUSSION

RFID automation delivers substantial operational improvements for small retail environments. The 1.3% remaining inventory inaccuracy stems from RFID tag detachment incidents. The reorder alert system effectively prevented critical stockouts. Future work should evaluate UHF RFID for longer read ranges and test scalability with 5,000+ SKU stores.